

Reproducibility Report for ‘A dual-pathway architecture for stress to disrupt agency and promote habit’ by Giovanniello et al. (2025, Nature)

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Introduction

[No abstract is needed.] Each reproducibility project will have a straightforward, no frills report of the study and reproducibility results. These reports will be publicly available as supplementary material for the aggregate report(s) of the project as a whole. Also, to maximize project integrity, the intro and methods will be written and critiqued in advance of your attempt to reproduce the results. Introductions can be just 1-2 paragraphs clarifying the main idea of the original study, the target finding for your reproducibility attempt, and any other essential information. It will NOT have a literature review – that is in the original publication. You can write both the introduction and the methods in past tense.

Giovanniello et al. (2025) investigated the neural mechanisms by which chronic stress disrupts agency learning to potentiate habit formation. In particular, the researchers investigated how distinct nuclei of the amygdala and their projections to the dorsal medial striatum support goal-directed vs habitual behavior, and the disruption of these processes by chronic stress. Chronic, mild stress perturbs these brain regions, rendering a mouse unable to exert volitional control over actions, as measured by two gold-standard assessments of goal-directed behavior: outcome devaluation and contingency degradation.

The focus of this report is to reproduce and simulate data to assess the authors’ core argument that stress disrupts goal-directed action and promotes habits, as indexed by insensitivity to contingency degradation. This behavioral finding is reflected by a Stress x Contingency degradation group interaction on lever press rates.

Clarify key analysis of interest here You can also pre-specify additional analyses you plan to do.

The main statistical analysis I will use to reproduce and simulate the main finding is a two-way ANOVA with stress and contingency degradation groups as factors. Lever-press rate during the contingency degradation probe test is what we will use as our measure.

Justification for choice of study

Please describe why you chose to reproduce the results of this study.

The researchers demonstrate the existence of and functional role for CeA-DMS afferents, a circuit I hope to investigate in the context of alcohol-induced aberrant motivational control over behavior. The DMS is a nucleus within the basal ganglia that has a role in the acquisition of goal-directed behavior that supports adaptive decision-making. The CeA is a nucleus within the amygdalar complex that has become a focal point of research on incentive salience. These regions, when investigated on their own, are compromised in disease states such as ethanol dependence, and investigating amygdalar-striatal afferents will provide knowledge on neural mechanisms of disrupted motivational and cognitive control over decision-making.

Anticipated challenges

Do you anticipate running into any challenges when attempting to reproduce these result(s)? If so please, list them here.

A challenge in simulation is capturing the natural variability of animal behavior since performance will follow a normal distribution guided by the means. Therefore, with simulations we lose entropy. The authors use a statistical analysis software called GraphPad Prism. For the purposes of the course and this report, I will use Python packages to reproduce and analyze simulated data. This difference in analysis platforms will yield different values, but should be somewhat similar. There are also many experiments described in this paper. Therefore, I will only focus on one: the effects of chronic stress on a contingency degradation probe test.

Links

Project repository (on Github): <https://github.com/tristantd/giovannielloetal2025.git>

Original paper (as hosted in your repo): https://github.com/psyc-201/giovannielloetal2025/tree/main/original_paper

Link to Pre-Registration: <https://osf.io/registries/drafts/691e0a9f9ef45ebbd997c92b/review>

Methods

Description of the steps required to reproduce the results

Please describe all the steps necessary to reproduce the key result(s) of this study.

1. Identify key behavior measure of the experiment to reproduce Assess effects of chronic stress on action-outcome learning using lever press rate during degraded and non-degraded contingencies.
2. Extract descriptive stats for each mouse in each group (press rates during degradation test) I will use available data included with the published paper to reproduce key statistics on Python.
3. Reanalyze using two-way ANOVA, Stress x Contingency degradation group The primary confirmatory analysis will be a Stress x Contingency degradation group interaction. We should also see an effect of contingency degradation and no effect of stress. This corresponds to figure 1i of the original paper.

4. Compare reproduced result to original reported values (F statistic, p-values, effect sizes)

Description of the steps required to simulate and analyze simulated dataset

The hypothesis I want to confirm is that chronic stress will reduce sensitivity to the action-outcome contingency. This would suggest stress mice have disrupted goal-directed control. To evaluate the effects of chronic stress on goal-directed actions, I will simulate behavioral data for 29 mice (7 control, non-degraded; 7 control, degraded; 7 stress, non-degraded; and 8 stress, degraded). These are the exact sample sizes for each group reported in the original paper.

The parameter estimates I will use are the means, SDs, SEMs for each mouse's press rates. Data simulation follows a normal distribution. The design is a 2 (control vs stress) x 2 (degraded vs nondegraded) between subjects ANOVA.

1. Use the original group means and SDs to parameterize the simulated data.
2. Generate simulated behavioral data from a normal distribution for each group.
3. Run the same statistical test as original authors, two-way ANOVA on press rate.
4. Visualization + Interpretation Create a bar graph comparing original and simulated findings side-by-side. Confirm whether the simulated data reproduce insensitivity to contingency degradation in stress mice.

Differences from original study

Explicitly describe known differences in the analysis pipeline between the original paper and yours (e.g., computing environment). The goal, of course, is to minimize those differences, but differences may occur. Also, note whether such differences are anticipated to influence your ability to reproduce the original results.

The differences from the original study are that this will be a simulation by which no behavioral data will be collected. Additionally, this paper uses a statistical analysis software called GraphPad Prism. For the purposes of this course, I will reproduce and simulate, as well as analyze the data using Python. I will use the same analysis, two-way ANOVA, for the simulated data. These differences will potentially influence my ability to reproduce the original results. For example, the different statistical analysis softwares may result in different reported values.

Project Progress Check 1

Measure of success

Please describe the outcome measure for the success or failure of your reproduction and how this outcome will be computed.

The outcome measure for the success of this reproducibility project will be if I can show the same effect that the authors claim in their paper (Stress x Contingency degradation group interaction). This will be comparing my simulated data with the authors' real reported data. This will be computed using an ANOVA on the simulated data and comparing effect sizes.

Pipeline progress

Earlier in this report, you described the steps necessary to reproduce the key result(s) of this study. Please describe your progress on each of these steps (e.g., data preprocessing, model fitting, model evaluation).

Results

Data preparation

Data preparation following the analysis plan.

```
### Data Preparation

#### Load Relevant Libraries and Functions

#### Import data

#### Data exclusion / filtering

#### Prepare data for analysis - create columns etc.
```

```
# Reproduce original data here

# IMPORTS

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
from statsmodels.stats.anova import AnovaRM, anova_lm
from statsmodels.formula.api import ols
import pingouin as pg

# Figure 1h DATA LOADING
data_1h = pd.read_csv('1h.csv')

data_1h = data_1h.rename(columns={'Subject ID': 'Subject_ID',
                                'Previous Experience': 'Stress'}) # Renaming
to avoid spaces in column names

df_1h = data_1h.melt( # put into a df
    id_vars=['Subject_ID', 'Group', 'Stress'],
    value_vars=['Day 1', 'Day 2', 'Day 3', 'Day 4'],
    var_name='Day',
    value_name='PressRate')
```

```

)

df_1h['Day'] = df_1h['Day'].str.replace("Day ", "").astype(int)
df_1h['Group'] = df_1h['Group'].astype('category')

print(df_1h.head(28))

pg.mixed_anova(
    data=df_1h,
    dv='PressRate',
    within='Day',
    between='Group',
    subject='Subject_ID'
)

# Plot for Figure 1h

plt.figure()

# Individual mouse lines

ind_mouse_line = sns.lineplot(
    data=df_1h,
    x='Day',
    y='PressRate',
    hue='Group',
    estimator=None,
    units='Subject_ID',
    lw=0.7,
    alpha=0.35,
    legend=False
)

# Main lines
main_group = sns.lineplot(
    data=df_1h,
    x='Day',
    y='PressRate',
    hue='Group',
    errorbar='se',
    lw=2.5,
    marker='o'
)

plt.title('Figure 1h Reproduction')
plt.xlabel('Day')
plt.xticks([1, 2, 3, 4])

```

```

plt.yticks(range(0, 35, 5))
plt.ylabel('Press Rate')
plt.show()

# Figure 1h STATISTICAL ANALYSIS

rmanova = pg.mixed_anova(
    dv='PressRate',
    within='Day',
    between='Group',
    subject='Subject_ID',
    data=df_1h
)

print(rmanova)

import statsmodels.api as sm
from statsmodels.formula.api import ols

# Fit the model
model = ols('PressRate ~ C(Day) * C(Group)', data=df_1h).fit()

# Type III SS (like SPSS/JAMOVII/JASP)
rmanova = sm.stats.anova_lm(model, typ=3)

print(rmanova)

# Figure 1i DATA LOADING

data_li = pd.read_csv('li.csv')

data_li = data_li.rename(columns={'Subject ID': 'Subject_ID', 'Press Rate':
    'Press_Rate'}) # Rename columns

data_li = data_li.rename(columns={'Previous Experience':
    'Previous_Experience'})

data_li['Group'] = data_li['Group'].astype('category')

data_li['Previous_Experience'] =
data_li['Previous_Experience'].astype('category')

print(data_li.head(30))

#Fixing Figure 1i ANOVA DATA LOADING

```

```

data_li_anova = pd.read_csv('li_anova.csv')

data_li_anova = data_li_anova.rename(columns={'Subject ID': 'Subject_ID',
'Press Rate': 'Press_Rate'}) # Rename columns

data_li_anova = data_li_anova.rename(columns={'Previous Experience':
'Previous_Experience'})

data_li_anova['Group'] = data_li_anova['Group'].astype('category')

data_li_anova['Previous_Experience'] =
data_li_anova['Previous_Experience'].astype('category')

print(data_li_anova.head(30))

# Plot for Figure 1i

plt.figure(figsize=(10,10)) #make figure bigger

# Bar plot: mean ± SE
sns.barplot(
    data=data_li,
    x='Group',
    y='Press_Rate',
    errorbar=('se'),          # standard error
    capsize=0.1,
    color='skyblue'
)

# Individual values
sns.stripplot(
    data=data_li,
    x='Group',
    y='Press_Rate',
    jitter=True,
    color='gray',          # lighter color for individual points
    alpha=0.6,
    size=6
)

plt.title('Figure 1i Reproduction')
plt.xlabel('Group')
plt.ylabel('Press Rate')
plt.yticks(range(0, 35, 5))
plt.show()

# Figure 1i two way anova

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```

anova = pg.anova(
    data=data_li_anova,
    dv='Press_Rate',
    between=['Previous_Experience', 'Group']
)

print(anova)

# Simulation Code Here

# Simulate Figure 1h and 1i from Giovanniello et al. (Nature, 2025)

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import pingouin as pg
import statsmodels.formula.api as smf

# Set random seed
np.random.seed(50)

# PARAMETERS / MOUSE GROUPS AND SESSION

n_mice = {
    'control_non_deg': 7, # 7 mice in control, non-degraded group
    'control_deg': 7, # 7 mice in control, degraded group
    'stress_non_deg': 7, # 7 mice in stress, non-degraded group
    'stress_deg': 8 # 8 mice in stress, degraded group
}
sessions = np.arange(1, 5) # 4 training sessions

print(n_mice)
print(sessions)

# Group means ± SDs from paper

training_data = { # these are training session means and SDs from the paper's
raw data, I got these numbers from uploading
    # the raw data onto JASP
    1: {'control_deg': (2.140, 0.619), # Day 1 mean and SD for control,
degraded group from paper raw data
        'control_non_deg': (1.630, 0.755), # Day 1 mean and SD for control,
non-degraded group from paper raw data
        'stress_deg': (2.235, 0.666), # Day 1 mean and SD for stress, degraded
group from paper raw data
        'stress_non_deg': (2.626, 1.056)}, # Day 1 mean and SD for stress,

```

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non-degraded group from paper raw data
    2: {'control_deg': (5.727, 2.088), # Day 2 mean and SD for control,
degraded group from paper raw data
    'control_non_deg': (6.240, 3.677), # Day 2 mean and SD for control,
non-degraded group from paper raw data
    'stress_deg': (4.484, 2.296), # Day 2 mean and SD for stress, degraded
group from paper raw data
    'stress_non_deg': (4.899, 2.223)}, # Day 2 mean and SD for stress,
non-degraded group from paper raw data
    3: {'control_deg': (10.28, 2.460), # Day 3 mean and SD for control,
degraded group from paper raw data
    'control_non_deg': (9.104, 2.770), # Day 3 mean and SD for control,
non-degraded group from paper raw data
    'stress_deg': (10.20, 2.811), # Day 3 mean and SD for stress, degraded
group from paper raw data
    'stress_non_deg': (10.98, 3.392)}, # Day 3 mean and SD for stress,
non-degraded group from paper raw data
    4: {'control_deg': (15.37, 2.336), # Day 4 mean and SD for control,
degraded group from paper raw data
    'control_non_deg': (16.69, 5.146), # Day 4 mean and SD for control,
non-degraded group from paper raw data
    'stress_deg': (14.98, 4.992), # Day 4 mean and SD for stress, degraded
group from paper raw data
    'stress_non_deg': (17.38, 6.069)} # Day 4 mean and SD for stress, non-
degraded group from paper raw data
}

degradation_data = { # these are degradation session means and SDs from the
paper's raw data, I got these numbers from
    # uploading the raw data onto JASP
    'Control Non-degraded': {'mean': 15.371, 'sd': 5.401, 'n': 7},
    'Control Degraded': {'mean': 3.486, 'sd': 1.852, 'n': 7},
    'Stress Non-degraded': {'mean': 12.225, 'sd': 5.537, 'n': 7},
    'Stress Degraded': {'mean': 11.714, 'sd': 4.730, 'n': 8}
}

# SIMULATE TRAINING DATA (Fig. 1h)
# From the means  $\pm$  SDs

training_sim = [] # to hold training data from simulation

# Create stable subject IDs
subjects_by_group = {
    group: [f"{group}_{i}" for i in range(n)]
    for group, n in n_mice.items()
}

for session, vals in training_data.items():

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for group, (mean, sd) in vals.items():
    subjects = subjects_by_group[group]
    # Simulate press rates
    rates = np.random.normal(mean, sd, len(subjects))
    # Remove negative values (press rates can't be negative)
    rates = np.clip(rates, 0, None)

    stress = 'Control' if 'control' in group else 'Stress'
    contingency = 'Degraded' if ('deg' in group and 'non' not in group)
else 'Non-degraded'

    for subj, rate in zip(subjects, rates):
        training_sim.append({
            'Session': session,
            'Subject': subj,
            'PressRate': rate,
            'Stress': stress,
            'Contingency': contingency
        })

training_df = pd.DataFrame(training_sim)
print(training_df.head(40))

# Save training data to CSV
training_df.to_csv('output.csv', index=False)

# PLOT SIMULATED TRAINING DATA

plt.figure(figsize=(8, 6))
sns.lineplot(
    data=training_df,
    x='Session', y='PressRate',
    hue='Stress', style='Contingency',
    markers=True, errorbar=('se'),
)

plt.title("Instrumental Training")
plt.xticks([1, 2, 3, 4])
plt.yticks([0, 2, 4, 6, 8, 10, 12, 14, 16, 18])
plt.ylabel("Presses per Minute")
plt.show()

# REPEATED MEASURES ANOVA for Training Data
# Mixed ANOVA with Session as within-subjects factor and Stress/Contingency as
between-subjects factors

# Create a combined Group factor for mixed_anova (which only supports one

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between factor)
training_df['Group'] = training_df['Stress'] + ' ' +
training_df['Contingency']

anova = pg.mixed_anova(
    dv='PressRate',
    within='Session',
    between='Group',
    subject='Subject',
    data=training_df
)

print(anova)

# SIMULATE CONTINGENCY DEGRADATION DATA (Fig. 1i)
# From real means  $\pm$  SDs

degradation_sim = [] # to hold lever press degradation data from simulation

for group, params in degradation_data.items():
    mean, sd, n = params['mean'], params['sd'], params['n']
    rates = np.random.normal(mean, sd, n) # simulate press rates for
degradation groups
    # Remove negative values
    rates = np.clip(rates, 0, None)

    stress = 'Control' if 'Control' in group else 'Stress'
    contingency = 'Degraded' if 'Degraded' in group else 'Non-degraded'

    for i, r in enumerate(rates):
        degradation_sim.append({
            'Subject': f"{group.replace(' ', '_')}_{i}",
            'Group': group,
            'Stress': stress,
            'Contingency': contingency,
            'PressRate': r
        })

degradation_df = pd.DataFrame(degradation_sim) # create dataframe
print(degradation_df.head(20)) # sanity check

# Save degradation data to CSV for analysis on JASP
degradation_df.to_csv('degradation_output.csv', index=False)

# PLOT SIMULATED CONTINGENCY DEGRADATION DATA

plt.figure(figsize=(10, 10))
sns.barplot(

```

```

    data=degradation_df, x='Group', y='PressRate',
    capsize=.1, errorbar='se', color='black'
)

sns.stripplot(
    data=degradation_df, x='Group', y='PressRate',
    jitter=True, alpha=0.6, color='grey', size=5
)
plt.title("Contingency Degradation Test")
plt.ylabel("Presses per Minute")
plt.xticks(rotation=40)
plt.tight_layout()
plt.show()

# TWO-WAY ANOVA for Degradation Data
# Between-subjects factors: Stress and Contingency

# Contingency degradation two-way ANOVA
# Using Stress and Contingency as separate factors (not Group, which is
redundant)

deg_anova = pg.anova(
    data=degradation_df,
    dv='PressRate',
    between=['Stress', 'Contingency']
)

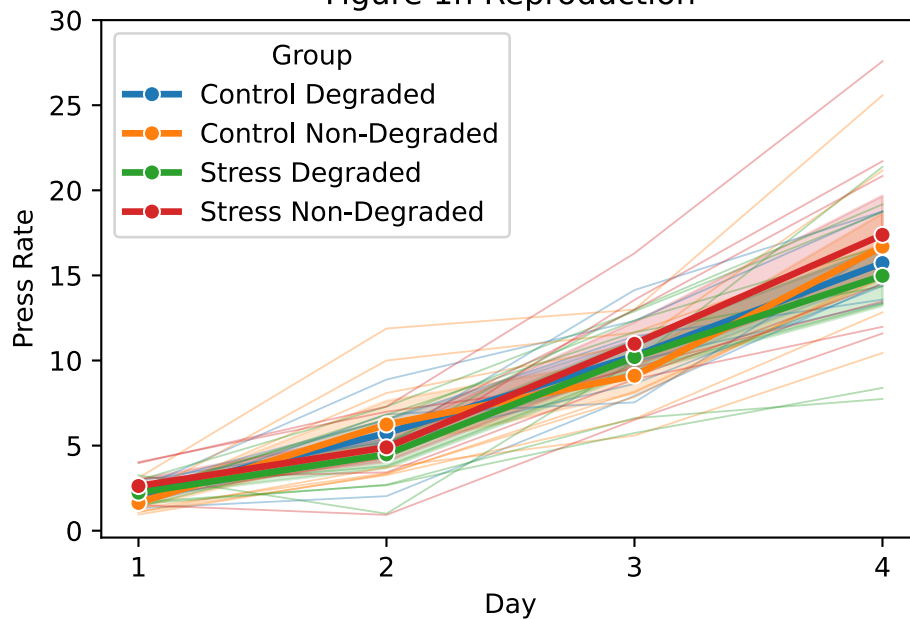
print(deg_anova)

```

	Subject_ID	Group	Stress	Day	PressRate
0	JG28 #17	Control Degraded	Handling	1	1.57
1	JG28 #18	Control Degraded	Handling	1	2.48
2	JG28 #19	Control Degraded	Handling	1	1.65
3	JG28 #20	Control Degraded	Handling	1	2.39
4	JG28 #21	Control Degraded	Handling	1	1.32
5	JG28 #22	Control Degraded	Handling	1	2.94
6	JG28 #23	Control Degraded	Handling	1	2.63
7	JG28 #25	Control Non-Degraded	Handling	1	1.97
8	JG28 #26	Control Non-Degraded	Handling	1	0.95
9	JG28 #27	Control Non-Degraded	Handling	1	3.10
10	JG28 #29	Control Non-Degraded	Handling	1	1.32
11	JG28 #30	Control Non-Degraded	Handling	1	1.85
12	JG28 #31	Control Non-Degraded	Handling	1	1.10
13	JG28 #32	Control Non-Degraded	Handling	1	1.12
14	JG28 #1	Stress Degraded	Stress	1	2.36
15	JG28 #2	Stress Degraded	Stress	1	3.26

16	JG28 #3	Stress Degraded	Stress	1	1.42
17	JG28 #4	Stress Degraded	Stress	1	2.96
18	JG28 #5	Stress Degraded	Stress	1	1.68
19	JG28 #6	Stress Degraded	Stress	1	1.57
20	JG28 #7	Stress Degraded	Stress	1	2.11
21	JG28 #8	Stress Degraded	Stress	1	2.52
22	JG28 #10	Stress Non-Degraded	Stress	1	3.97
23	JG28 #11	Stress Non-Degraded	Stress	1	1.92
24	JG28 #12	Stress Non-Degraded	Stress	1	1.90
25	JG28 #13	Stress Non-Degraded	Stress	1	3.08
26	JG28 #14	Stress Non-Degraded	Stress	1	4.03
27	JG28 #15	Stress Non-Degraded	Stress	1	1.97

Figure 1h Reproduction



	Source	SS	DF1	DF2	MS	F	p-unc
0	Group	14.910402	3	25	4.970134	0.207350	
8.903396e-01							
1	Day	3236.926166	3	75	1078.975389	210.845696	
1.859144e-36							
2	Interaction	40.154902	9	75	4.461656	0.871865	
5.539952e-01							
	p-GG-corr	np2	eps	sphericity	W-spher	p-spher	
0	NaN	0.024278	NaN	NaN	NaN	NaN	
1	1.281160e-24	0.893998	0.603633	False	0.363513	0.000057	
2	NaN	0.094714	NaN	NaN	NaN	NaN	

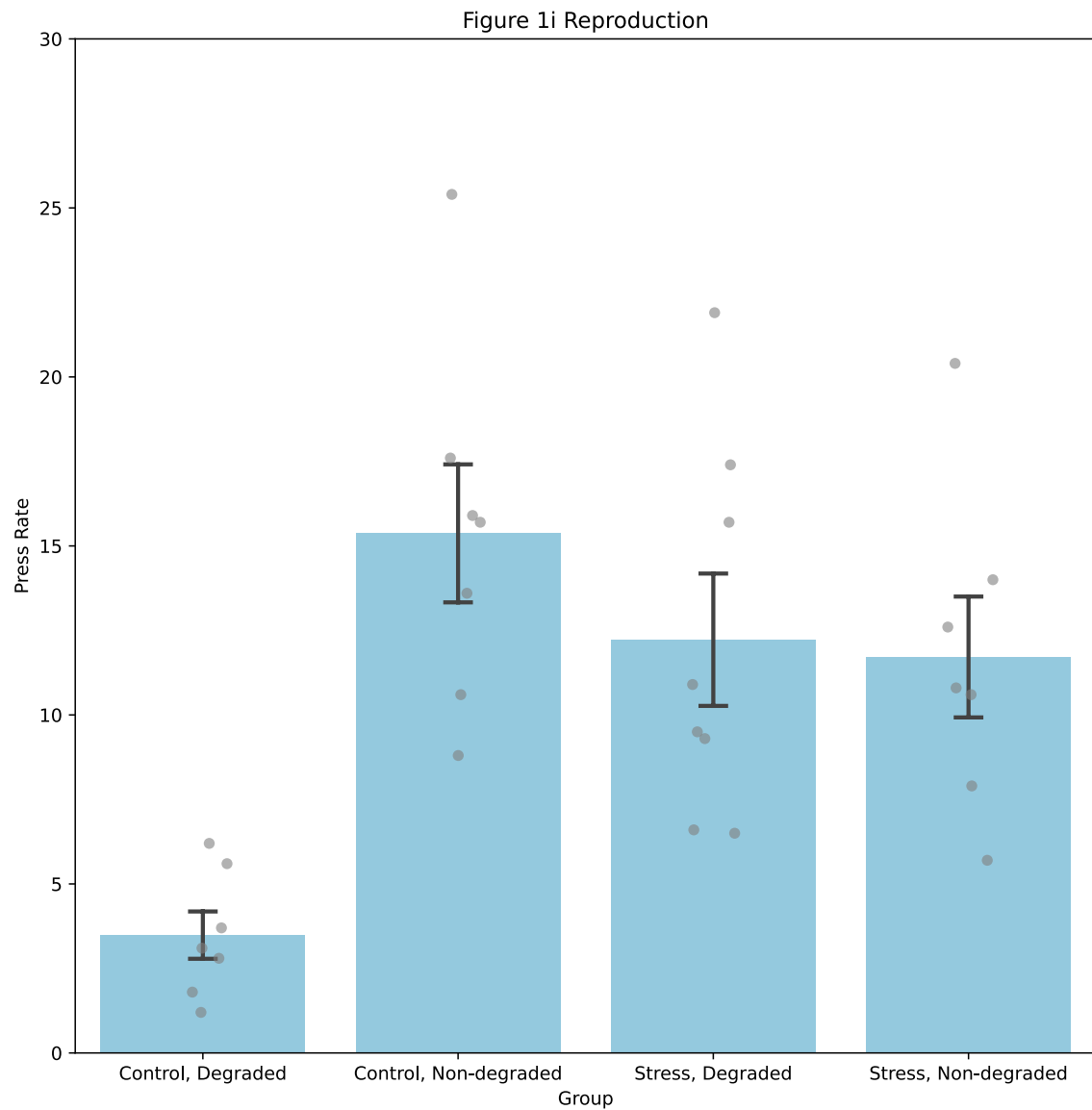
		sum_sq	df	F	PR(>F)
Intercept		32.057200	1.0	3.261007	7.395405e-02
C(Day)		724.486886	3.0	24.566054	5.431414e-12
C(Group)		3.532325	3.0	0.119775	9.482826e-01
C(Day):C(Group)		40.154902	9.0	0.453860	9.018246e-01
Residual		983.046075	100.0	NaN	NaN
Subject_ID	Sex	Age at Start	Strain	Previous_Experience	\
0	JG28 #17	M	9 weeks	C57/Bl6	Handling
1	JG28 #18	M	9 weeks	C57/Bl6	Handling
2	JG28 #19	M	9 weeks	C57/Bl6	Handling
3	JG28 #20	M	9 weeks	C57/Bl6	Handling
4	JG28 #21	F	9 weeks	C57/Bl6	Handling
5	JG28 #22	F	9 weeks	C57/Bl6	Handling
6	JG28 #23	F	9 weeks	C57/Bl6	Handling
7	JG28 #25	M	9 weeks	C57/Bl6	Handling
8	JG28 #26	M	9 weeks	C57/Bl6	Handling
9	JG28 #27	M	9 weeks	C57/Bl6	Handling
10	JG28 #29	F	9 weeks	C57/Bl6	Handling
11	JG28 #30	F	9 weeks	C57/Bl6	Handling
12	JG28 #31	F	9 weeks	C57/Bl6	Handling
13	JG28 #32	F	9 weeks	C57/Bl6	Handling
14	JG28 #1	M	9 weeks	C57/Bl6	Stress
15	JG28 #2	M	9 weeks	C57/Bl6	Stress
16	JG28 #3	M	9 weeks	C57/Bl6	Stress
17	JG28 #4	M	9 weeks	C57/Bl6	Stress
18	JG28 #5	F	9 weeks	C57/Bl6	Stress
19	JG28 #6	F	9 weeks	C57/Bl6	Stress
20	JG28 #7	F	9 weeks	C57/Bl6	Stress
21	JG28 #8	F	9 weeks	C57/Bl6	Stress
22	JG28 #10	M	9 weeks	C57/Bl6	Stress
23	JG28 #11	M	9 weeks	C57/Bl6	Stress
24	JG28 #12	M	9 weeks	C57/Bl6	Stress
25	JG28 #13	F	9 weeks	C57/Bl6	Stress
26	JG28 #14	F	9 weeks	C57/Bl6	Stress
27	JG28 #15	F	9 weeks	C57/Bl6	Stress
28	JG28 #16	F	9 weeks	C57/Bl6	Stress
Group	Press_Rate				
0	Control, Degraded		3.1		
1	Control, Degraded		5.6		
2	Control, Degraded		1.2		
3	Control, Degraded		6.2		
4	Control, Degraded		1.8		
5	Control, Degraded		2.8		
6	Control, Degraded		3.7		
7	Control, Non-degraded		17.6		
8	Control, Non-degraded		15.9		
9	Control, Non-degraded		25.4		

10	Control, Non-degraded	13.6					
11	Control, Non-degraded	10.6					
12	Control, Non-degraded	15.7					
13	Control, Non-degraded	8.8					
14	Stress, Degraded	9.5					
15	Stress, Degraded	21.9					
16	Stress, Degraded	15.7					
17	Stress, Degraded	17.4					
18	Stress, Degraded	9.3					
19	Stress, Degraded	6.6					
20	Stress, Degraded	10.9					
21	Stress, Degraded	6.5					
22	Stress, Non-degraded	20.4					
23	Stress, Non-degraded	14.0					
24	Stress, Non-degraded	12.6					
25	Stress, Non-degraded	10.6					
26	Stress, Non-degraded	10.8					
27	Stress, Non-degraded	5.7					
28	Stress, Non-degraded	7.9					
	Subject_ID	Sex	Age	at Start	Strain	Previous_Experience	Group \
0	JG28 #17	M	9 weeks	C57/Bl6	Handling	Degraded	
1	JG28 #18	M	9 weeks	C57/Bl6	Handling	Degraded	
2	JG28 #19	M	9 weeks	C57/Bl6	Handling	Degraded	
3	JG28 #20	M	9 weeks	C57/Bl6	Handling	Degraded	
4	JG28 #21	F	9 weeks	C57/Bl6	Handling	Degraded	
5	JG28 #22	F	9 weeks	C57/Bl6	Handling	Degraded	
6	JG28 #23	F	9 weeks	C57/Bl6	Handling	Degraded	
7	JG28 #25	M	9 weeks	C57/Bl6	Handling	Non-Degraded	
8	JG28 #26	M	9 weeks	C57/Bl6	Handling	Non-Degraded	
9	JG28 #27	M	9 weeks	C57/Bl6	Handling	Non-Degraded	
10	JG28 #29	F	9 weeks	C57/Bl6	Handling	Non-Degraded	
11	JG28 #30	F	9 weeks	C57/Bl6	Handling	Non-Degraded	
12	JG28 #31	F	9 weeks	C57/Bl6	Handling	Non-Degraded	
13	JG28 #32	F	9 weeks	C57/Bl6	Handling	Non-Degraded	
14	JG28 #1	M	9 weeks	C57/Bl6	Stress	Degraded	
15	JG28 #2	M	9 weeks	C57/Bl6	Stress	Degraded	
16	JG28 #3	M	9 weeks	C57/Bl6	Stress	Degraded	
17	JG28 #4	M	9 weeks	C57/Bl6	Stress	Degraded	
18	JG28 #5	F	9 weeks	C57/Bl6	Stress	Degraded	
19	JG28 #6	F	9 weeks	C57/Bl6	Stress	Degraded	
20	JG28 #7	F	9 weeks	C57/Bl6	Stress	Degraded	
21	JG28 #8	F	9 weeks	C57/Bl6	Stress	Degraded	
22	JG28 #10	M	9 weeks	C57/Bl6	Stress	Non-Degraded	
23	JG28 #11	M	9 weeks	C57/Bl6	Stress	Non-Degraded	
24	JG28 #12	M	9 weeks	C57/Bl6	Stress	Non-Degraded	
25	JG28 #13	F	9 weeks	C57/Bl6	Stress	Non-Degraded	
26	JG28 #14	F	9 weeks	C57/Bl6	Stress	Non-Degraded	
27	JG28 #15	F	9 weeks	C57/Bl6	Stress	Non-Degraded	

28	JG28 #16	F	9 weeks	C57/Bl6	Stress	Non-Degraded
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	Press_Rate					
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0	3.1					
1	5.6					
2	1.2					
3	6.2					
4	1.8					
5	2.8					
6	3.7					
7	17.6					
8	15.9					
9	25.4					
10	13.6					
11	10.6					
12	15.7					
13	8.8					
14	9.5					
15	21.9					
16	15.7					
17	17.4					
18	9.3					
19	6.6					
20	10.9					
21	6.5					
22	20.4					
23	14.0					
24	12.6					
25	10.6					
26	10.8					
27	5.7					
28	7.9					

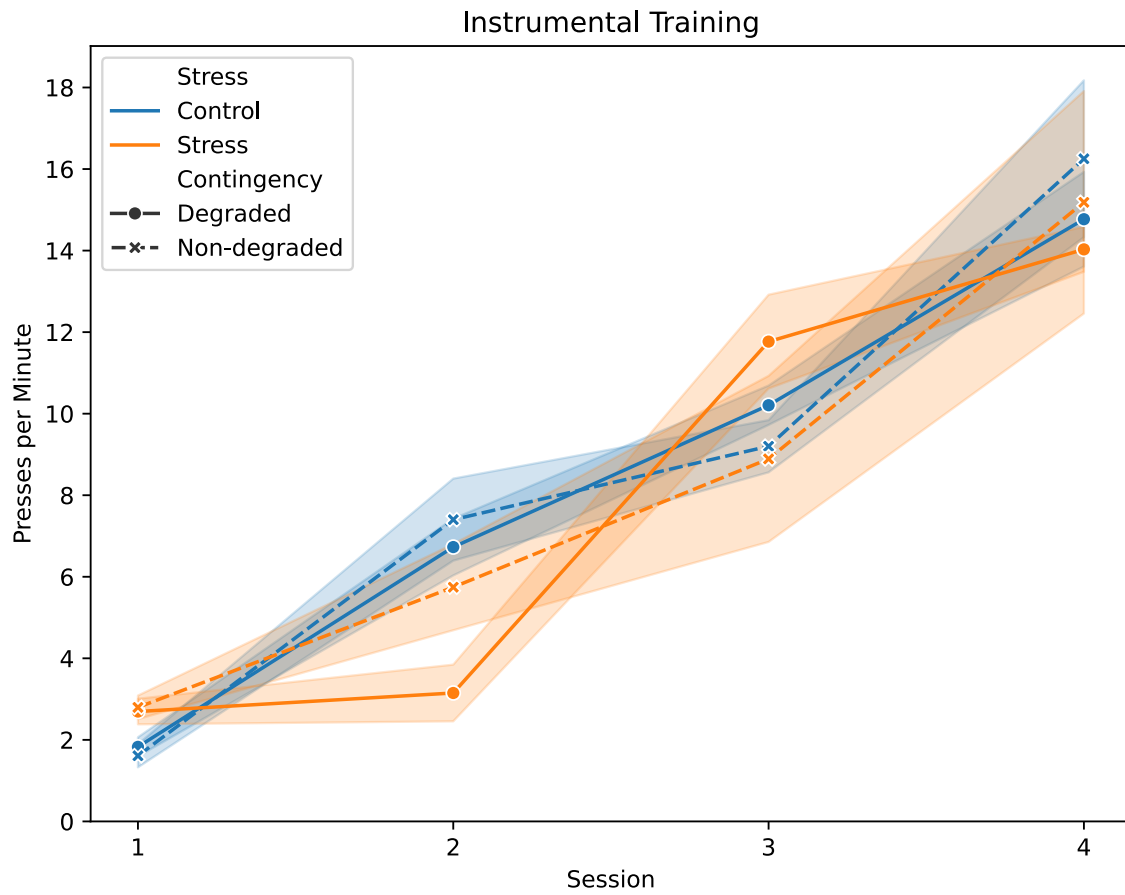


	Source	SS	DF	MS	F \
0	Previous_Experience	54.345167	1.0	54.345167	2.495432
1	Group	217.819453	1.0	217.819453	10.001877
2	Previous_Experience * Group	277.600023	1.0	277.600023	12.746893
3	Residual	544.446429	25.0	21.777857	NaN

	p-unc	np2
0	0.126748	0.090758
1	0.004072	0.285753
2	0.001480	0.337694
3	NaN	NaN

{'control_non_deg': 7, 'control_deg': 7, 'stress_non_deg': 7, 'stress_deg': 8}

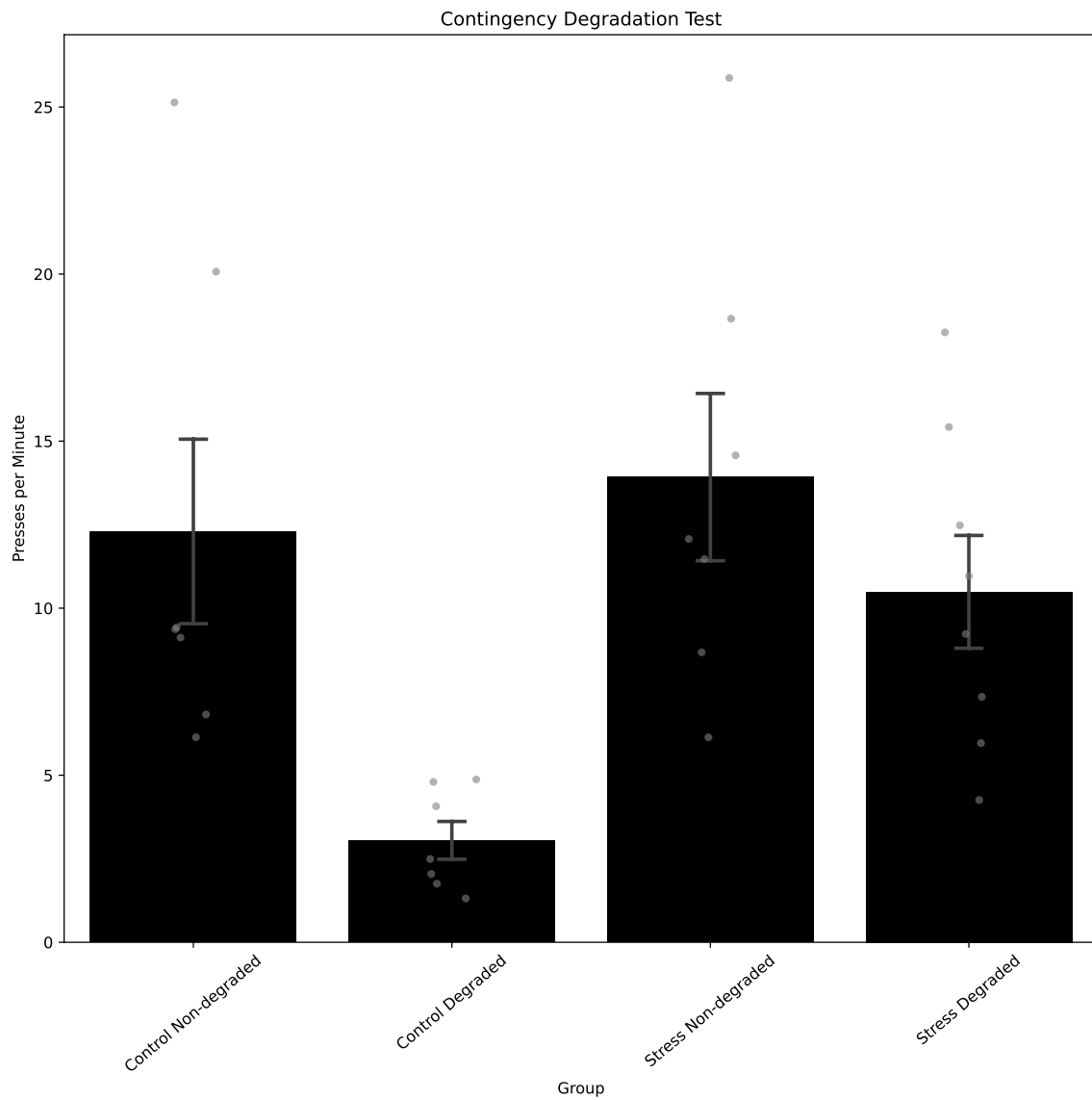
[1 2 3 4]					
	Session	Subject	PressRate	Stress	Contingency
0	1	control_deg_0	1.174142	Control	Degraded
1	1	control_deg_1	2.120825	Control	Degraded
2	1	control_deg_2	1.755645	Control	Degraded
3	1	control_deg_3	1.233425	Control	Degraded
4	1	control_deg_4	3.013995	Control	Degraded
5	1	control_deg_5	1.844903	Control	Degraded
6	1	control_deg_6	1.656890	Control	Degraded
7	1	control_non_deg_0	2.438052	Control	Non-degraded
8	1	control_non_deg_1	0.661869	Control	Non-degraded
9	1	control_non_deg_2	0.627753	Control	Non-degraded
10	1	control_non_deg_3	1.725385	Control	Non-degraded
11	1	control_non_deg_4	2.280956	Control	Non-degraded
12	1	control_non_deg_5	2.156036	Control	Non-degraded
13	1	control_non_deg_6	1.377403	Control	Non-degraded
14	1	stress_deg_0	1.570648	Stress	Degraded
15	1	stress_deg_1	3.299873	Stress	Degraded
16	1	stress_deg_2	4.442174	Stress	Degraded
17	1	stress_deg_3	2.892855	Stress	Degraded
18	1	stress_deg_4	2.317495	Stress	Degraded
19	1	stress_deg_5	2.729695	Stress	Degraded
20	1	stress_deg_6	1.972625	Stress	Degraded
21	1	stress_deg_7	2.333645	Stress	Degraded
22	1	stress_non_deg_0	2.190680	Stress	Non-degraded
23	1	stress_non_deg_1	2.456285	Stress	Non-degraded
24	1	stress_non_deg_2	2.773345	Stress	Non-degraded
25	1	stress_non_deg_3	2.927456	Stress	Non-degraded
26	1	stress_non_deg_4	2.328987	Stress	Non-degraded
27	1	stress_non_deg_5	4.432718	Stress	Non-degraded
28	1	stress_non_deg_6	2.467846	Stress	Non-degraded
29	2	control_deg_0	7.168360	Control	Degraded
30	2	control_deg_1	8.013797	Control	Degraded
31	2	control_deg_2	8.521597	Control	Degraded
32	2	control_deg_3	2.868566	Control	Degraded
33	2	control_deg_4	6.742661	Control	Degraded
34	2	control_deg_5	7.300353	Control	Degraded
35	2	control_deg_6	6.485914	Control	Degraded
36	2	control_non_deg_0	5.082810	Control	Non-degraded
37	2	control_non_deg_1	11.289555	Control	Non-degraded
38	2	control_non_deg_2	3.944018	Control	Non-degraded
39	2	control_non_deg_3	7.621647	Control	Non-degraded



	Source	SS	DF1	DF2	MS	F	p-unc	\
0	Group	8.188973	3	25	2.729658	0.347850	7.909933e-01	
1	Session	2665.349360	3	75	888.449787	86.297506	2.949194e-24	
2	Interaction	136.548640	9	75	15.172071	1.473704	1.733743e-01	
	p-GG-corr	np2	eps	sphericity	W-spher	p-spher		\
0	NaN	0.040069	NaN	NaN	NaN	NaN		
1	1.011927e-18	0.775377	0.72606	False	0.384348	0.000111		
2	NaN	0.150270	NaN	NaN	NaN	NaN		
	Subject	Group	Stress	Contingency	\			
0	Control_Non-degraded_0	Control	Non-degraded	Control	Non-degraded			
1	Control_Non-degraded_1	Control	Non-degraded	Control	Non-degraded			
2	Control_Non-degraded_2	Control	Non-degraded	Control	Non-degraded			
3	Control_Non-degraded_3	Control	Non-degraded	Control	Non-degraded			
4	Control_Non-degraded_4	Control	Non-degraded	Control	Non-degraded			
5	Control_Non-degraded_5	Control	Non-degraded	Control	Non-degraded			
6	Control_Non-degraded_6	Control	Non-degraded	Control	Non-degraded			
7	Control_Degraded_0	Control	Degraded	Control	Degraded			
8	Control_Degraded_1	Control	Degraded	Control	Degraded			

9	Control_Degraded_2	Control Degraded	Control	Degraded
10	Control_Degraded_3	Control Degraded	Control	Degraded
11	Control_Degraded_4	Control Degraded	Control	Degraded
12	Control_Degraded_5	Control Degraded	Control	Degraded
13	Control_Degraded_6	Control Degraded	Control	Degraded
14	Stress_Non-degraded_0	Stress Non-degraded	Stress	Non-degraded
15	Stress_Non-degraded_1	Stress Non-degraded	Stress	Non-degraded
16	Stress_Non-degraded_2	Stress Non-degraded	Stress	Non-degraded
17	Stress_Non-degraded_3	Stress Non-degraded	Stress	Non-degraded
18	Stress_Non-degraded_4	Stress Non-degraded	Stress	Non-degraded
19	Stress_Non-degraded_5	Stress Non-degraded	Stress	Non-degraded

	PressRate
0	6.136125
1	6.817603
2	9.427255
3	9.120652
4	25.136250
5	9.369524
6	20.072113
7	4.873098
8	4.800054
9	4.071509
10	2.044747
11	2.493399
12	1.312063
13	1.754757
14	6.134080
15	12.069998
16	8.677876
17	14.573965
18	18.663771
19	25.870229



	Source	SS	DF	MS	F	p-unc	\
0	Stress	154.795871	1.0	154.795871	5.115640	0.032660	
1	Contingency	282.246118	1.0	282.246118	9.327572	0.005300	
2	Stress * Contingency	61.052117	1.0	61.052117	2.017629	0.167834	
3	Residual	756.483344	25.0	30.259334	NaN	NaN	
np2							
0		0.169867					
1		0.271722					
2		0.074678					
3		NaN					

Key analysis

The analyses as specified in the analysis plan.

Analyses for reproduced original data and simulation data are in code block above. We cannot input only the analyses and figures here because they require all lines of code (e.g., imports, csv, etc).

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Reproduction Attempt

Open the discussion section with a paragraph summarizing the primary result from the key analysis and assess whether you successfully reproduced it, partially reproduced it, or failed to reproduce it.

Giovanniello et al. (2025) report that chronic, mild stress disrupts action-outcome learning and potentiates habit formation, reflected by a Stress x Contingency degradation group interaction $F(1,25) = 12.75$, $p = 0.002$. I was able to reproduce this statistical result from the available data, $F(1,25) = 12.75$, $p < 0.002$, $\eta^2 = 0.34$. Using simulated data based on the same group statistics and sample sizes, I observed the interaction, $F(1,25) = 34.59$, $p < 0.001$, $\eta^2 = 0.81$. Although the exact values differed, performance patterns were comparable. The exact values differed due to the normal distribution imposed during the simulation, but the behavioral pattern was consistent with the original. This indicates a successful reproduction of the authors' argument—stress shifts behavior away from goal-directed control toward habit-like responding. Because of this simulation and low entropy, the effect size I observed is quite large.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis of the dataset, (b) assessment of the meaning of the successful or unsuccessful reproducibility attempt - e.g., for a failure to reproduce the original findings, are the differences between original and present analyses ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the reproducibility attempt (if you contacted them). None of these need to be long.

Although the reproducibility attempt was successful, it is important to consider that there are limitations to simulation-based analyses and that these approaches simplify the biological context. Real animal behavior typically shows heterogeneity driven by motivation, stress exposure variability, reinforcement history, and learning rate differences. Simulating data from normal distributions will reduce noise and inflate effect sizes, overestimating group separability.

Additionally, the real experiment was embedded within a chronic stress protocol that may introduce compounding physiological factors not captured in simulated data. Nonetheless, I was

able to reproduce the key finding of stress-induced impairment to adaptive decision-making. This supports the original conclusion and provides confidence in using CeA-DMS circuitry as a target for future investigations, especially in my field, whereby alcohol dependence disrupts motivational control. For reproducibility projects, you should also answer the following four questions – these are for the interpretation/conclusion portion, which otherwise doesn't have the same content as the replication project.

1. What assumptions does your generative model make about participant behavior, and which aspects of the real experiment does it simplify or omit? This simulation project assumes normally distributed behavior within groups, consistent instrumental response acquisition within conditions, and uniform effects of stress. This simulation omits motivational processes, physiological consequences of chronic stress (HPA axis regulation), and learning dynamics that shape the behavioral trajectories (i.e., sustained press rate across days).
2. Show how your simulated dataset flows through your planned analysis pipeline. Does this recover any mismatches or bugs? Upon data simulation, I needed to adjust the columns for different factors in accordance with Python packages to analyze that data. Essentially, each factor needed its own column. The original paper did not have to do this. They used GraphPad Prism in which they could copy and paste their values onto a sheet. My pre-processing resolved bugs when attempting to use analysis packages in Python. My analysis pipeline is aligned with the logic of the original paper, though extra steps were necessary.
3. Identify one insight your simulation gave you about the strengths or limitations of the original experimental design. This simulation highlighted to me that variability in behavior can control the effect we can observe. Although I used the SDs from the original paper, my simulation was constrained to a normal distribution, causing the interaction effect to be larger.
4. How would this simulation help you design a follow-up experiment with a similar paradigm? The simulation estimates effect magnitude and variance that can help with experimental design (e.g., power calculation and sample size). It suggests that future experiments need to incorporate individual difference measurements to include variability that is absent in the model. Follow-up experimental designs could incorporate session-level models to assess changes in learning strategy across sessions, and this can improve sensitivity to subtle shifts in goal-directed control.